

Natnael Alemayehu

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WORK EXPERIENCE

TReNDS Center, Atlanta, Georgia

Aug 2025 – Present

Graduate Research Assistant

- Applying Topological Data Analysis to study brain connectivity differences between schizophrenia & bipolar disorder.
- Developing Python-based supervised ML pipelines to analyze fMRI data from 800+ subjects, using GPU-accelerated HPC clusters and Slurm-managed parallel workflows to identify topological biomarkers of psychiatric disorders.

Microsoft, Atlanta, Georgia

May 2025 – July 2025

Software Engineer Intern (Azure Core - High Performance Compute / AI Team)

- Designed and deployed an end-to-end diagnostic data pipeline in Azure Synapse and Python to process logs from thousands of GPU nodes, automating issue detection and eliminating manual querying.
- Built interactive Power BI dashboards that visualized system faults and performance trends, speeding up root-cause analysis by 50%.
- Developed a Python-based real time K-means clustering model to group similar error patterns from real-time logs, improving fault classification and guiding scalable diagnostics.

Hence Technologies, London, United Kingdom (Remote)

June 2021 – Aug 2024

Software Engineer

- Developed a contract management platform using React, Redux, Node.js with Express, and MongoDB with advanced search through Elasticsearch integration, real-time annotations, and role-specific views
- Designed GraphQL APIs with resolver batching, pagination, and Redis caching, reducing over-fetching and cutting average client query times by 30%.
- Implemented secure multi-tenant authentication using JWT with refresh tokens, middleware-based RBAC, and encrypted session storage, ensuring compliance with enterprise security standards.
- Optimized MongoDB schema and queries with compound indexes and aggregation pipelines, increasing ingestion throughput 4x while reducing query response times under heavy load.
- Built CI/CD pipelines with GitHub Actions and Docker, deploying containerized services to AWS ECS with blue/green rollout strategies, reducing release cycles from days to under 24 hours.
- Migrated codebase for a claims management application from React to Next.js, reducing page-load time by 25%.
- Developed a Google Chrome extension using Next.js to streamline lawyer selection and evaluation.

PROJECTS

LightWeight CNN for classification of Brain Tumors from MRI Images, [Paper Link](#)

Jan 2025 - May 2025

- Developed a light weight CNN network to classify MRI brain scans into multiple tumor types, applying preprocessing techniques such as normalization, augmentation, and ROI cropping; paper cited within 3 months of publication.

Real-time Ethiopian Sign Language Detection system with LSTM Neural Network

Feb 2022 – May 2022

- Built and deployed an LSTM neural network model as a web app using Flask that is capable of detecting Ethiopian sign language in real time with an accuracy score of 83%.

Participant, Google Software Product Sprint (Google SPS) Program

May 2021 – Aug 2021

- Developed a location based food recommendation web application using React, Node, and Tailwind CSS.

SKILLS

Technical Skills (Proficient): React, Next.js, Node.js, Python, Flask, PostgreSQL, SQL, MongoDB, Azure, AWS, Docker, Kubernetes, Airflow, Spark, TensorFlow, PyTorch, Git, CI/CD, Data Modeling, Visualization (Power BI, Tableau)

EDUCATION

Georgia State University, Atlanta, Georgia (CGPA: 4.0 / 4.0)

Expected graduation: Dec 2025

Area of Study: Master of Science, Computer Science; Interest Areas: Data Mining, Artificial Intelligence, Computer Vision

African Leadership University (ALU), Kigali, Rwanda

Aug 2018 – July 2022

- Area of Study: Bachelor of Science (Hons) Computer Science
- Awards/Honors: Full Scholarship, Top Data and Decisions Course Student Award